Advanced Modeling

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Necessary Packages

```
library(tidyverse)
library(rsample)
library(caret)
library(vip)
library(ROCR)
library(nnet)
```

Necessary Data Sets

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
teamvisdata <- readRDS('Data-NBA/Current/teamvisdata.rds')
teamvisdata_fixed <- na.omit(teamvisdata) %>%
    mutate(
        team = as.factor(team),
        opponent = as.factor(opponent),
        season = as.factor(season))
```

Linear Regression Model Testing: Player Statistics

Logistic Regression Model Testing: Binary Game Outcome

```
winbinary_split <- initial_split(teamvisdata_fixed, prop=.75, strata='twoleveloc')
winbinary_train <- training(winbinary_split)
winbinary_test <- testing(winbinary_split)
```

For the models I started by adding all the variables that I wanted and deciding which ones I wanted to remove to see how the accuracy changes.

```
winbinary_1 <- train(
    twoleveloc ~ roll_offrating + roll_defrating + team + homeaway +
        vorp_bpa + roll_three_par + roll_ftar + roll_ast_pct,
    data=teamvisdata_fixed,
    method='glm',
    family='binomial',
    trControl=trainControl(method='cv', number=10))
winbinary_2 <- train(
    twoleveloc ~ roll_offrating + roll_defrating,
    data=teamvisdata_fixed,
```

```
    method='glm',
    family='binomial',
    trControl=trainControl(method='cv', number=10))
winbinary_3 <- train(
    twoleveloc ~ roll_offrating + opponent,
    data=teamvisdata_fixed,
    method='glm',
    family='binomial',
    trControl=trainControl(method='cv', number=10))
winbinary_4 <- train(
    twoleveloc ~ homeaway,
    data=teamvisdata_fixed,
    method='glm',
    family='binomial',
    trControl=trainControl(method='cv', number=10))
```

This will give the me summary statistics of the models' accuracy.

```
summary(
    resamples(
        list(
            model1=winbinary_1,
            model2=winbinary_2,
            model3=winbinary_3,
            model4=winbinary_4
        )
    )
)$statistics$Accuracy
```

```
##             Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## model1 0.6500333 0.6613440 0.6628955 0.6634959 0.6665003 0.6790945    0
## model2 0.6373919 0.6531936 0.6576846 0.6580372 0.6622853 0.6852961    0
## model3 0.6054558 0.6162050 0.6264138 0.6272298 0.6409913 0.6447106    0
## model4 0.5355955 0.5507319 0.5547425 0.5545662 0.5612109 0.5739015    0
```

ROC Curve for Logistic Regression Models

Creating a plot to show the False Positive vs. True Positive Accuracy.

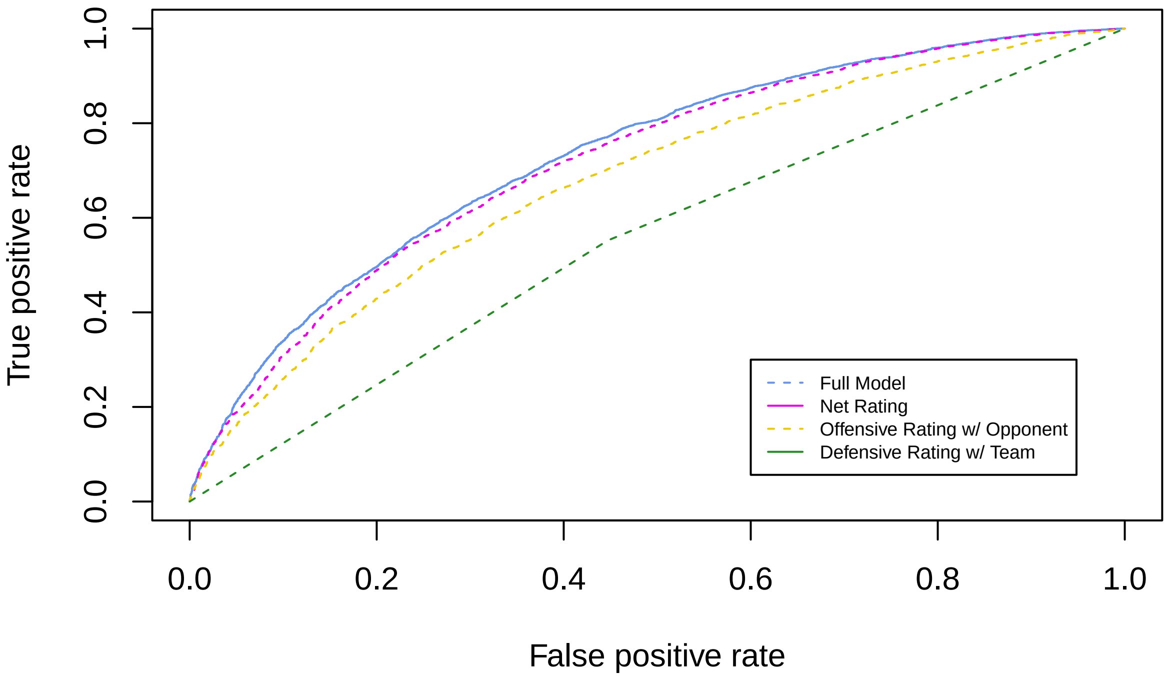
```
model1_prob <- predict(winbinary_1, winbinary_train, type='prob')$Win
perf1 <- prediction(model1_prob, winbinary_train$twoleveloc) %>%
    performance(measure='tpr', x.measure='fpr')

model2_prob <- predict(winbinary_2, winbinary_train, type='prob')$Win
perf2 <- prediction(model2_prob, winbinary_train$twoleveloc) %>%
    performance(measure='tpr', x.measure='fpr')

model3_prob <- predict(winbinary_3, winbinary_train, type='prob')$Win
perf3 <- prediction(model3_prob, winbinary_train$twoleveloc) %>%
    performance(measure='tpr', x.measure='fpr')

model4_prob <- predict(winbinary_4, winbinary_train, type='prob')$Win
perf4 <- prediction(model4_prob, winbinary_train$twoleveloc) %>%
    performance(measure='tpr', x.measure='fpr')
```

```r
plot(perf1, col='cornflowerblue', lty=1)
plot(perf2, add=TRUE, col='magenta2', lty=2)
plot(perf3, add=TRUE, col='gold2', lty=2)
plot(perf4, add=TRUE, col='forestgreen', lty=2)
legend(0.6, 0.3, legend=c('Full Model',
                          'Net Rating',
                          'Offensive Rating w/ Opponent',
                         'Defensive Rating w/ Team'),
       col=c('cornflowerblue', 'magenta2', 'gold2', 'forestgreen'),
       lty=2:1, cex=0.6)
```



Multinomial Regression Testing: Outcome w/ Magnitude

```r
wincat_split <- initial_split(teamvisdata_fixed, prop=0.75, strata = 'fourleveloc')
wincat_train <- training(wincat_split)
wincat_test <- testing(wincat_split)
```

```r
wincat_1 <- train(
    fourleveloc ~ roll_offrating + roll_defrating + team + homeaway +
        vorp_bpa + roll_three_par + roll_ftar + roll_ast_pct,
    data=teamvisdata_fixed,
    method='multinom',
    trace=FALSE)

wincat_2 <- train(
    fourleveloc ~ homeaway,
    data=teamvisdata_fixed,
    method='multinom',
    trace=FALSE)
```

```r
summary(
    resamples(
        list(
            model1=wincat_1,
            model2=wincat_2
        )
    )
)$statistics$Accuracy
```

```
##             Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## model1 0.3701837 0.3764579 0.3777617 0.3792042 0.3829361 0.3899565    0
## model2 0.2656760 0.2779888 0.2820235 0.2815202 0.2865731 0.2908345    0
```